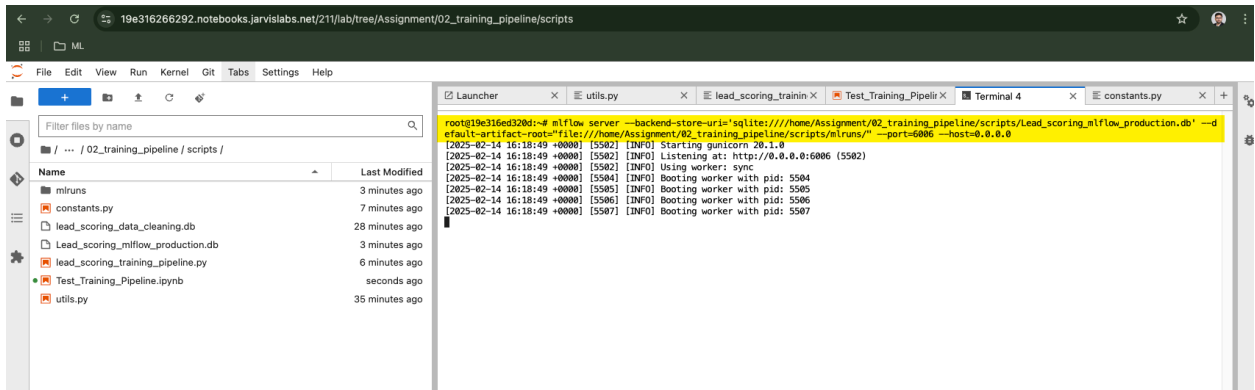


ML OPS Training Pipeline

Running the ML Flow Server



The screenshot shows a JupyterLab interface with a file browser on the left and a terminal window on the right. The file browser displays the contents of the `02_training_pipeline/scripts` directory, including files like `mlruns`, `constants.py`, `lead_scoring_data_cleaning.db`, `Lead_scoring_mlflow_production.db`, `lead_scoring_training_pipeline.py`, `Test_Training_Pipeline.ipynb`, and `utils.py`. The terminal window shows the command `mlflow server --backend-store-uri=sqlite:///home/Assignment/02_training_pipeline/scripts/Lead_scoring_mlflow_production.db --default-artifact-root=file:///home/Assignment/02_training_pipeline/scripts/mlruns/ --port=6006 --host=0.0.0.0` being executed, followed by logs indicating the server is starting and listening on port 6006.

```
root@19e316ed320d:~# mlflow server --backend-store-uri=sqlite:///home/Assignment/02_training_pipeline/scripts/Lead_scoring_mlflow_production.db --d
efault-artifact-root=file:///home/Assignment/02_training_pipeline/scripts/mlruns/ --port=6006 --host=0.0.0.0
[2025-02-14 16:18:49 +0000] [5502] [INFO] Starting gunicorn 20.1.0
[2025-02-14 16:18:49 +0000] [5502] [INFO] Listening at: http://0.0.0.0:6006 (5502)
[2025-02-14 16:18:49 +0000] [5502] [INFO] Using worker: sync
[2025-02-14 16:18:49 +0000] [5504] [INFO] Booting worker with pid: 5504
[2025-02-14 16:18:49 +0000] [5505] [INFO] Booting worker with pid: 5505
[2025-02-14 16:18:49 +0000] [5506] [INFO] Booting worker with pid: 5506
[2025-02-14 16:18:49 +0000] [5507] [INFO] Booting worker with pid: 5507
```

Experiments - Lead_scoring_mlflow_production

19e3162662921.notebooks.jarvislabs.net/#/experiments/2

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Lead_scoring_mlflow_production

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Experiment ID: 2

Description [Edit](#)

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Showing 1 matching run

	Start Time	Duration	Run Name	User	Source	Version	Models	Metrics	Parameters
☐	21 seconds ago	8.8s	run_LightGB	root	ipykerne...	-	LightGBM4	False Negative: 6190, Precision: 0.739, Precision_0: 0.789	boosting_type: gbd, class_weight: None

[Load more](#)

19e3162662921.notebooks.jarvislabs.net/#/experiments/2/runs/93fe3ca12aa648519d5c5e735a91a479

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Lead_scoring_mlflow_production > run_LightGB

run_LightGB

Date: 2025-02-14 22:01:06 Source: ipykernel_launcher.py User: root

Duration: 8.8s Status: FINISHED Lifecycle Stage: active

Description [Edit](#)

Parameters (20)

Metrics (12)

Tags

Artifacts

models

- MLmodel
- conda.yaml
- model.pkl
- python_env.yaml
- requirements.txt

Full Path: file:///home/Assignment/02_training_pipeline/scripts/mlruns/2/93fe3ca12aa648519d5c5e735a91a479/artifacts/models

LightGBM v4 Registered on 2025/02/14

MLflow Model

The code snippets below demonstrate how to make predictions using the logged model. This model is also registered to the [model registry](#).

Model schema

Input and output schema for your model. [Learn more](#)

Name	Type
No schema. See MLflow docs for how to include input and output schema with your model.	

Make Predictions

Predict on a Spark DataFrame:

```
import mlflow
logged_model = 'runs:/93fe3ca12aa648519d5c5e735a91a479/models'

# Load model as a Spark UDF. Override result_type if the model does not return double values.
loaded_model = mlflow.pyfunc.spark_udf(spark, model_uri=logged_model, result_type='double')

# Predict on a Spark DataFrame.
columns = list(df.columns)
df.withColumn('predictions', loaded_model(*columns)).collect()
```

19e3162662921.notebooks.jarvislabs.net/#/models/LightGBM

mlflow 1.26.1 Experiments Models GitHub Docs

Registered Models > LightGBM

LightGBM

Created Time: 2025-02-14 21:49:18 Last Modified: 2025-02-14 22:17:18

► Description [Edit](#)

► Tags

▼ Versions All Active 1 Compare

☐	Version	Registered at	Created by	Stage	Description
☐	Version 4	2025-02-14 22:01:13		Production	
☐	Version 3	2025-02-14 21:59:14		None	
☐	Version 2	2025-02-14 21:49:42		None	
☐	Version 1	2025-02-14 21:49:18		None	

< 1 >

19e3162662921.notebooks.jarvislabs.net/#/experiments/2/runs/93fe3ca12aa648519d5c5e735a91a479

mlflow 1.26.1 Experiments Models GitHub Docs

Lead_scoring_mlflow_production > run_LightGB

run_LightGB

Date: 2025-02-14 22:01:06 Source: `ipykernel_launcher.py` User: root

Duration: 8.8s Status: FINISHED Lifecycle Stage: active

► Description [Edit](#)

► Parameters (20)

▼ Metrics (12)

Name	Value
False Negative 🔗	6190
Precision 🔗	0.739
Precision_0 🔗	0.789
Precision_1 🔗	0.706
Recall 🔗	0.747
Recall_0 🔗	0.649
Recall_1 🔗	0.829
True Negative 🔗	23095
auc 🔗	0.747
f1_0 🔗	0.712
f1_1 🔗	0.762